



SURGICAL PATHOLOGY REPORT

Breast: wide local excision with sentinel lymph node biopsy

FINAL REPORT

1 PATIENT & CASE DETAILS

PATIENT NAME Sunita Rajesh Verma	MRN / UHID ONC-2026-004217	AGE 52 years
SEX ASSIGNED AT BIRTH Female	DATE OF BIRTH 14-Mar-1974	MENOPAUSAL STATUS Postmenopausal
ACCESSION NO. SP-26-08731	REPORT STATUS Final	REFERRING CLINICIAN Dr. Anil Deshmukh, MS, MCh (Surgical Oncology)
REFERRING UNIT Department of Surgical Oncology	DATE OF PROCEDURE 22-Sep-2026	DATE RECEIVED 22-Sep-2026, 14:10 IST
DATE REPORTED 29-Sep-2026, 17:35 IST	NEOADJUVANT THERAPY None	

FINAL DIAGNOSIS

A. RIGHT BREAST, WIDE LOCAL EXCISION:

- Invasive carcinoma of no special type (ductal NST), Nottingham grade 2 (score 7/9: tubule 3, nuclear 2, mitoses 2)
- Tumour size: 3.4 cm (greatest dimension); upper outer quadrant
- Lymphovascular invasion: **present (focal)**
- Associated ductal carcinoma in situ (DCIS), intermediate nuclear grade, cribriform and solid types
- Margins: **negative** for invasive carcinoma and DCIS (closest: invasive carcinoma 0.4 cm, deep margin)

B. RIGHT AXILLARY SENTINEL LYMPH NODES (3):

- Metastatic carcinoma in 2 of 3 sentinel nodes (largest deposit 4.5 mm); no extranodal extension

PATHOLOGIC STAGE (AJCC 8th ed.): pT2 pN1a(sn) cM0, anatomic stage group IIB

BIOMARKERS: ER positive (90%, strong) | PR positive (70%, moderate) | HER2 negative (IHC 1+, HER2-low) | Ki-67 12% | TILs 15% | Oncotype DX RS 19 | MammaPrint low risk | BRCA1/BRCA2 germline negative

2 CLINICAL HISTORY & CLINICAL NOTES

History: 52-year-old postmenopausal woman with a palpable lump in the upper outer quadrant of the right breast, first detected on screening mammography (BI-RADS 5). Ultrasound-guided core biopsy (outside centre) showed invasive carcinoma; the axilla was clinically and sonographically node-negative, so sentinel lymph node biopsy was planned. Family history of breast cancer in a maternal aunt. No prior breast surgery. No neoadjuvant therapy.

Clinical notes (from requisition): Firm, non-tender, mobile lump of approximately 3.5 cm in the right upper outer quadrant. No skin tethering, peau d'orange, or nipple retraction. No palpable axillary or supraclavicular nodes. No clinical or radiological evidence of distant metastasis (CT chest/abdomen and bone scan: negative). Comorbidities: type 2 diabetes mellitus, hypertension, and osteoporosis, all on treatment. Documented penicillin allergy (rash).

3 SPECIMEN & GROSS DESCRIPTION

Part	Specimen	Gross findings
A	Right breast, wide local excision (short suture superior, long suture lateral, double suture deep). Weight 74 g; 8.5 × 6.0 × 3.8 cm.	All margins inked. Serial sectioning shows a firm, stellate, grey-white tumour measuring 3.4 × 2.9 × 2.6 cm in the upper outer quadrant. Closest margin: deep, 0.4 cm. All other margins 1.0 cm or more. Surrounding breast is fibrofatty. Entire lesion and all margins submitted (cassettes A1 to A14).

Part	Specimen	Gross findings
B	Right axillary sentinel lymph nodes (3 nodes; radioisotope and blue-dye positive).	B1: 1.6 cm, tan-white with firm tan area. B2: 1.2 cm, tan-white. B3: 0.7 cm, fatty hilum. All nodes bisected or serially sliced and entirely submitted (B1 to B6).

4 MICROSCOPIC DESCRIPTION

Sections show an invasive carcinoma composed of irregular nests, cords, and occasional poorly formed glands infiltrating a desmoplastic stroma with an infiltrative growth pattern. Tumour cells have moderately pleomorphic vesicular nuclei with visible nucleoli. Tumour necrosis is absent. Focal lymphovascular invasion is identified in peritumoural lymphatic spaces. Perineural invasion is not identified. Stromal tumour-infiltrating lymphocytes are present at approximately 15%. Adjacent to the invasive tumour there is ductal carcinoma in situ of intermediate nuclear grade with cribriform and solid architecture, without comedo necrosis, extending over approximately 1.1 cm. The nipple and skin are not included in the specimen. Background breast shows usual ductal hyperplasia and fibrocystic change.

Nottingham (Elston-Ellis) component	Finding	Score
Tubule / gland formation	Less than 10% of tumour area forms glands	3
Nuclear pleomorphism	Moderate variation in size, shape, and chromatin	2
Mitotic count	8 mitoses per 10 high-power fields (field diameter 0.50 mm)	2
Total score / Overall grade	Moderately differentiated	7 / Grade 2

5 MARGIN STATUS & REGIONAL LYMPH NODES

Margin	Distance to invasive carcinoma	Distance to DCIS	Status
Deep (posterior)	0.4 cm	0.9 cm	Negative
Superior	1.2 cm	0.6 cm	Negative
Inferior / Medial / Lateral / Anterior	1.0 cm or more	1.0 cm or more	Negative

Node	Size	Result	Largest deposit	Extranodal extension
B1	1.6 cm	Metastatic carcinoma	4.5 mm (macrometastasis)	Absent
B2	1.2 cm	Metastatic carcinoma	2.8 mm (macrometastasis)	Absent
B3	0.7 cm	No tumour identified	None	Not applicable
Total	3 nodes	2 of 3 positive	Isolated tumour cells: none	No ENE in any node

6 PATHOLOGIC STAGING (AJCC CANCER STAGING MANUAL, 8TH EDITION)

Category	Value	Criteria
pT: primary tumour	pT2	Tumour greater than 2 cm but not more than 5 cm (3.4 cm)
pN: regional nodes	pN1a(sn)	Metastases in 1 to 3 axillary nodes, at least one deposit larger than 2 mm (2 of 3 sentinel nodes; 4.5 mm)
M: distant metastasis	cM0	No clinical or radiological evidence of distant metastasis
Anatomic stage group	Stage IIB	T2 N1 M0

7 IMMUNOHISTOCHEMISTRY: BIOMARKER PANEL

Block A7 (representative invasive tumour). Fixation: 10% neutral buffered formalin for 24 hours; cold ischaemia time under 60 minutes. Positive and negative external controls and internal normal-duct controls: satisfactory. Scoring per ASCO/CAP guidelines.

Biomarker	Clone	Result	Quantification	Interpretation
Estrogen receptor (ER)	SP1	POSITIVE	90% of tumour nuclei; strong (3+); Allred 8/8 (PS 5 + IS 3)	ER positive (1% or more nuclear staining)
Progesterone receptor (PR)	1E2	POSITIVE	70% of tumour nuclei; moderate (2+); Allred 7/8 (PS 5 + IS 2)	PR positive; high expression (20% or more)
HER2 (IHC)	4B5	NEGATIVE	Score 1+: incomplete, faint membrane staining in more than 10% of tumour cells	HER2 negative (HER2-low); ISH not required
Ki-67 proliferation index	30-9	12%	Counted in 500 invasive tumour cells (international Ki-67 working group method)	Low proliferation (below 14% cut-off)
Tumour-infiltrating lymphocytes (TILs)	H&E	15%	Stromal TILs, assessed per International TILs Working Group	Intermediate stromal TIL density

8 GENOMIC & GERMLINE ASSAYS (ORDERED SEPARATELY)

Performed at an external reference laboratory; reports on file and summarised here for completeness.

Assay	Result	Category / Interpretation
BRCA1 (germline multigene panel)	NEGATIVE	No pathogenic or likely pathogenic variant detected
BRCA2 (germline multigene panel)	NEGATIVE	No pathogenic or likely pathogenic variant detected
Oncotype DX Recurrence Score	19	Intermediate risk (range 11 to 25; TAILORx categories)
MammaPrint (70-gene signature)	LOW RISK	MammaPrint index +0.142; low risk of distant recurrence

9 SYSTEMIC HEALTH PROFILE (FROM CLINICAL RECORD & PRE-OPERATIVE WORK-UP)

Parameter	Value	Details
LVEF (cardiac function)	58%	2D transthoracic echocardiography, 15-Sep-2026. Normal systolic function (normal range 55% or more); no regional wall motion abnormality.
ECOG performance status	1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature.

Comorbidity	Present	Details
Cardiac disease	No	No history of coronary disease, heart failure, arrhythmia, or prior cardiotoxic therapy
Diabetes mellitus	YES	Type 2 diabetes mellitus; well controlled on metformin (HbA1c 6.8%)
Hypertension	YES	Essential hypertension; controlled on amlodipine (clinic BP 128/80 mmHg)
Osteoporosis	YES	Lumbar spine DEXA T-score -2.6 (Aug 2026); on alendronate with calcium and vitamin D

Current medication	Dose & frequency	Indication
Metformin	500 mg orally, twice daily	Type 2 diabetes mellitus
Amlodipine	5 mg orally, once daily	Hypertension
Alendronate	70 mg orally, once weekly	Osteoporosis
Calcium carbonate + Vitamin D3	500 mg + 1000 IU orally, once daily	Bone health

Allergies: Penicillin (skin rash). No other known drug allergies.

10 SURROGATE MOLECULAR CLASSIFICATION & PATHOLOGIST'S COMMENT

IHC-based surrogate subtype: Luminal A-like (ER positive, PR positive at 20% or more, HER2 negative, Ki-67 below 14%; St. Gallen 2013 surrogate definitions). This is a surrogate classification from immunohistochemistry only.

Comment: Grade 2 invasive carcinoma of no special type, ER/PR positive and HER2 negative (HER2-low), measuring 3.4 cm with focal lymphovascular invasion and macrometastases in 2 of 3 sentinel lymph nodes (pN1a(sn)). All surgical margins are negative for invasive carcinoma and DCIS; the deep margin is the closest at 0.4 cm. Genomic risk assays are discordant (Oncotype DX recurrence score 19, intermediate; MammaPrint low risk), and nodal involvement is present; these should be interpreted together with clinical risk factors. Discussion at the multidisciplinary tumour board is recommended to determine adjuvant systemic, endocrine, and radiation therapy and the need for further axillary management.

Electronically signed by Dr. Meera Iyer, MD (Pathology) Consultant Pathologist, Breast & Oncologic Pathology Reg. No.: SAMPLE-000000 Signed: 29-Sep-2026, 17:35 IST	Grossing & IHC Grossed by: Dr. Rohan Patil, Pathology Resident IHC performed: 24-Sep-2026 to 25-Sep-2026 IHC reviewed by: Dr. Meera Iyer Genomic reports verified: 28-Sep-2026
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11 STRUCTURED DATA SUMMARY (SYNOPTIC FIELDS)

Consolidated values for the patient, tumour, biomarker, and health-profile fields of this case, to support structured data entry and automated extraction.

Field	Value	Source in report
PATIENT		
Full name	Sunita Rajesh Verma	Header
Age	52 years	Header
Sex assigned at birth	Female	Header
Clinical notes	Firm mobile 3.5 cm right UOQ lump; no skin/nipple changes; axilla clinically negative	Section 2
TUMOUR & STAGING		
Clinical / anatomic stage	II (Stage IIB; pT2 pN1a(sn) cM0)	Section 6
Histological grade	2 (Nottingham score 7/9)	Section 4
Tumour size	3.4 cm	Section 3 / 4
Lymph node involvement	Positive; positive node count = 2 (of 3 sentinel nodes)	Section 5
BIOMARKERS		
ER status	Positive (90%, Allred 8/8)	Section 7
PR status	Positive (70%, Allred 7/8)	Section 7
HER2 status	Negative (IHC 1+, HER2-low)	Section 7
Ki-67	12%	Section 7
BRCA1 / BRCA2	Negative / Negative (germline)	Section 8
TILs	15%	Section 7
Oncotype DX score	19 (intermediate risk)	Section 8
MammaPrint	Low risk	Section 8
SYSTEMIC HEALTH PROFILE		
LVEF	58%	Section 9
ECOG score	1	Section 9
Comorbidities	Diabetes, Hypertension, Osteoporosis (Cardiac: none)	Section 9
Medications	Metformin, Amlodipine, Alendronate, Calcium + Vitamin D3	Section 9
Allergies	Penicillin (rash)	Section 9 / 2

References: AJCC Cancer Staging Manual, 8th ed.; Elston-Ellis (Nottingham) histologic grading; ASCO/CAP guidelines for ER/PR and HER2 testing in breast cancer; International TILs Working Group recommendations; St. Gallen International Consensus (2013) surrogate subtype definitions; TAILORx recurrence-score categories. **End of report.**